Required A2 Tests: 2d and 3d Lattice Dirac Pass the Locked Gate; the 1d Scalar Instrument is Rejected

Robert W. McGwier
CTO, Cohere Technology Group
GitHub: `n4hy`

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Abstract

The same pre-registered locality gate as Paper 21: $R(m)/R(0) < 2$ for $0 < mL \leq 8$, local ansatz = on-site + nearest neighbours.

2d staggered fermion, 32×32 grid, 12×12 region: ratios $1.19, 1.29, 1.38, 1.38$. Mutation (drop diagonal) fires. Auditor on $N = 28$, $L = 10$: 1.35, 1.29. C2 PASS. [**P**] as a necessary-condition test.

3d staggered fermion, 10^3 grid, 6^3 region: ratios $0.45, 0.47, 0.44$. C2 PASS. Mass shortens the modular range.

1d massive scalar: no verdict. Symplectic eigenvalues sat on the clip $\xi = 1/2 + \varepsilon$ and $R_\pi(0) = 0.95$. The instrument is rejected. That is not a physics fail of A2.

A2 is not refuted on free lattice Dirac in $d = 1, 2, 3$. It is not a 4d continuum theorem and not the Standard Model.

1 Gate

Unchanged from M9.10. A spacetime scalar is on-site. Anything beyond the lattice adjacency cannot be δX.

2 2d

$R(0) = 0.118$. Inside the window the ratio stays ≤ 1.38. The auditor, different size, confirms. The remainder still grows with m, as in 1d. It does not cross 2.

3 3d

$R(0) = 0.300$ (small region). Finite mass *decreases* R. C2 passes. Finite-volume caveat is recorded.

4 Scalar

The bosonic reconstruction produced ξ_{min} pinned at the floor and a massless K_π that is 95% nonlocal. A massless harmonic interval should be CHM-local in the continuum. The run is discarded.

Admission. *I ran the tests I said were required. I did not promote a broken scalar kernel into a fail, and I did not write that Jacobson 2016 is now a 4d theorem.*